Natural Computing
https://doi.org/10.1007/s11047-025-10040-x

Mycelium as a computational medium: a framework for growth modeling towards reservoir computing

Ioannis Tompris[1] · Ioannis K. Chatzipaschalis[1,2] · Theodoros Panagiotis Chatzinikolaou[1] · Georgios Kleitsiotis[1] · Karolos-Alexandros Tsakalos[1] · Iosif-Angelos Fyrigos[1] · Michail-Antisthenis Tsompanas[3] · Andrew Adamatzky[3] · Phil Ayres[4] · Georgios Ch. Sirakoulis[1]

Accepted: 2 July 2025
© The Author(s) 2025

Abstract
Mycelium, the intricate vegetative network of fungi, has emerged as a promising candidate within the realm of engineered living materials (ELMs). While its intriguing structural and electrical properties highlight its potential, mycelium growth is highly sensitive to environmental conditions. To bridge this gap, a robust framework was developed to both model mycelium growth and explore its computational capabilities. This framework uses a cellular automata (CA) approach, enhanced with reaction-diffusion (RD) processes, to simulate mycelium growth under diverse environmental conditions. This configuration, combined with tunable parameters, enables the identification and validation of optimal growth patterns, supported by an algorithm designed to extract key features of hyphae–the fundamental building blocks of the mycelial network. Subsequently, the small-world properties of the modeled mycelium networks were investigated, revealing high clustering coefficients and short path lengths, characteristics that make them well-suited for reservoir computing (RC). To demonstrate their computational capabilities, mycelium-inspired RC architectures were evaluated on the MNIST dataset classification task, achieving an accuracy of up to 97.09%, highlighting the effectiveness of biologically inspired models. As a result, this framework establishes a comprehensive test-bench for mycelium modeling, growth, and computational exploration, paving the way for innovative applications in bio-inspired computing.

Keywords Mycelium · Reaction diffusion · Cellular automata · Electrical activity · Small-world · Reservoir computing · Neuron

1 Introduction

Engineered Living Materials (ELMs) are considered as an emerging field of research, characterized by their unique ability to integrate self-healing, regenerative, and adaptive traits of biological systems within contemporary materials (Mora-Boza et al. 2023). Among various ELMs, mycelium-based materials are particularly notable (Elsacker et al. 2023; Karana et al. 2018; Adamatzky 2023) and serve as the focus of this study. This is because mycelium, the vegetative structure of fungi, is widely recognized for its growth and sustainability, offering an eco-friendly alternative to conventional materials while being abundantly available in nature (Ayres et al. 2022; Angelova et al. 2021). Furthermore, mycelium exhibits electrical activity, recently explored as a potential computing approach. More specifically, some studies (Roberts and Adamatzky 2023; Dehshibi and Adamatzky 2021; Adamatzky et al. 2019) demonstrate the distinctive electrical behavior of mycelium in response to various electrical stimuli, revealing a complex, brain-like spiking pattern that encapsulates the depolarization, repolarization, and refractory phases characteristic of biological spikes, together with low-voltage and sparse spiking activity.

By integrating these electrical properties, combined with mycelium's structural characteristics, which are indicative of small-world topologies, we enter the domain of liquid state machines (LSMs), a class of networks that utilize bio-inspired architectures and intricate dynamics (Tsakalos et al. 2021; Beiler et al. 2010; Dale et al. 2021; Maass 2011). LSMs belong to the broader framework of reservoir computing (RC) (Tanaka et al. 2019), which is utilized for solving pattern recognition problems and cognitive tasks. In this approach,

Extended author information available on the last page of the article

Published online: 13 August 2025

input data is mapped onto a high-dimensional space representation through the reservoir dynamics, allowing for the extraction of temporal and spatial features. The reservoir, often composed of spiking neurons or non-linear dynamical units, uses recurrent connections and temporal memory to process sequential data, making it suitable for time-dependent tasks, such as pattern recognition and classification. LSMs can recognize complex patterns over time, making them useful for tasks that require adaptability.

Despite its promising topology, electrical activity, and computing potential, mycelium faces challenges as a functional material. One of the primary obstacles lies in its sensitivity to environmental conditions, which necessitates highly controlled cultivation environments. Precise regulation of temperature, humidity, and light is essential to maintain optimal growth and functionality, making large-scale production and experimentation both resource-intensive and technically demanding (Omuse et al. 2022; Arya and Singh 2016; Jesse 2023; Lin et al. 2021). Additionally, the commercialization of mycelium-based materials is hindered by stringent regulatory frameworks (Aiduang et al. 2024; Molitorisová et al. 2021; Molitorisová and Monaco 2023). As a result, while mycelium holds immense potential as a sustainable and computationally active material, its practical implementation remains a challenging endeavor.

In this work, we exploit the unique topology and electrical activity of mycelium to implement biologically inspired reservoir networks for RC. By utilizing the mycelium's small-world network properties and brain-like spiking behavior, we demonstrate effective RC performance, achieving up to 97.09% accuracy on the MNIST classification task. In this direction, a proof-of-concept framework, outlined in Fig. 1, is proposed. Section 2 introduces a reaction-diffusion *Mycelium Model* within a Cellular Automata (CA) framework, incorporating adjustable *Fitting Parameters & Environmental Conditions* to simulate mycelium growth under various factors. Section 3 details the *Hyphae Information* algorithm, which evaluates growth and extracts *Small-World Metrics* to assess network topology. Section 4 implements *Reservoir Computing* utilizing this topology, with configurable *Training Options* and *Network Evaluation* methods, which include both contemporary and biologically-inspired cost functions. Finally, Section 5 concludes the paper providing a thorough outline of its most important aspects.

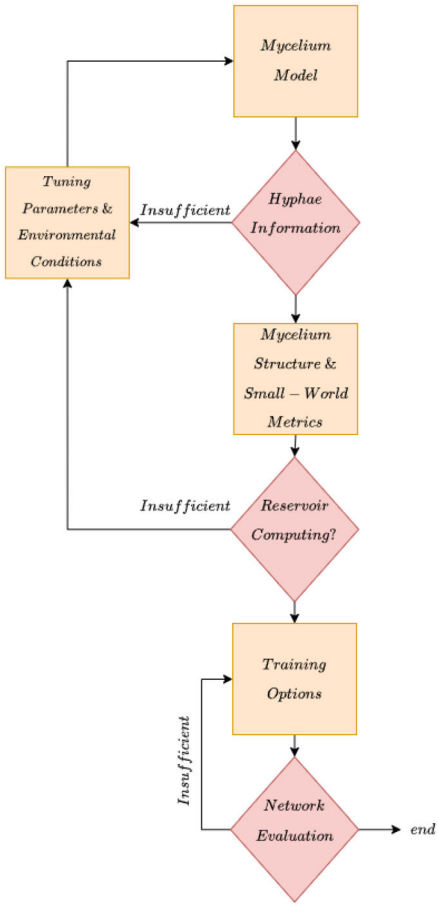


Fig. 1 State machine of the proposed framework

2 Mycelium model

Reaction-Diffusion (RD) processes are essential in this model as they enable accurate fungal growth simulations, including mechanisms such as hyphal extension, anastomosis, and apical and lateral branching (Fricker et al. 2017; Boswell 2008; Tompris et al. 2024). The model is developed within a CA framework, which provides a discrete, lattice-based structure that enhances parallelism and allows precise control over the diffusion dynamics of the ELM, (Tsompanas et al. 2022; Pavlidis et al. 2023; Chatzinikolaou et al. 2024; Vaxevanellis et al. 2024) ensuring seamless integration with the desired RD processes (Codd 2014; Weimar 1997). In contrast, lattice-free approaches documented in the literature demonstrate promising modeling capabilities (Carver and Boswell 2008; Vidal-Diez de Ulzurrun et al. 2017), although they often incur increased computational complexity and lack the inherent parallelism offered by CA-based system. Notably, the proposed model emerges as an ideal candidate for implementation on efficient parallel hardware (Ntinas et al. 2020; Karamani et al. 2021; Chatzinikolaou et al. 2022; Chatzipaschalis et al. 2023), facilitating the development of a real-time digital twin (Chatzipaschalis et al. 2024).

2.1 Reaction-diffusion processes

The mycelium model employed is grounded in RD processes (Sugimura et al. 2007; Tompris et al. 2024), as articulated in Eqs. (1) - (4). It describes the growth of

mycelium, represented by the concentration c, as the outcome of intricate interactions between the activator and the inhibitor, components u and v respectively, with the former promoting mycelium growth, while the latter denotes it.

$$\frac{\partial u}{\partial t} = \nabla^2 u + \zeta(\kappa u + u^2 - \lambda uv)n, \quad u \in \Omega_c \tag{1}$$

$$\frac{\partial v}{\partial t} = d\nabla^2 v + \zeta(\mu u^3 - v), \quad v \in \Omega \tag{2}$$

$$\frac{dc}{dt} = \zeta vc(a(u) - c)(c - 1), \quad c \in \Omega \tag{3}$$

$$a(u) = \begin{cases} \xi, & \text{if } u \geq \text{threshold} \\ \xi - \theta(u - \text{threshold}), & \text{otherwise} \end{cases} \tag{4}$$

A critical aspect to consider is the initial segment of Eqs. (1) and (2), which relates to the diffusion term. This segment employs the Laplacian operator ∇ to precisely describe the spatial propagation of these components, with the parameter d governing the rate at which the inhibitor expands relative to the activator. The second term in the corresponding equations, a polynomial of u and v is called the reaction term and denotes the interaction between the two components and their respective effects on each other. Herein, the term ζ linearly scales the values of u and v to desirable ranges, while the adjustable terms d, κ, λ and μ are treated as fitting parameters and allow for various growth patterns, explained thoroughly in Section 3.1.

The variable spaces of Ω and Ω_c and the differences between them should also be noted. While Ω represents growth of a component throughout the whole grid, Ω_c, tied to the activator, defines a growth that is restricted within a specified radius relative to c, effectively replicating the plasma membrane of a biological organism and maintaining biological fidelity. Finally, moving on to Eqs. (3) and (4), it is evident that the activator u indirectly regulates the growth of mycelium c through the function $a(u)$, indicating that growth occurs only when the activator reaches or exceeds a specific threshold value. Lastly, in Eq. (4), the values of the parameters ξ and θ, depend on the defined threshold and enable c to assume both positive and negative values, through $a(u)$.

2.2 Environmental conditions

Mycelial growth can be affected by environmental factors such as temperature, humidity, and light. In the model, the influence of these environmental conditions on mycelial growth is computed on the factor n of Eq. (1), which affects the reaction term of the activator, encapsulating the aggregated environmental effects, as shown in Eq. (5), thereby dynamically modifying growth behaviors and environmental interactions.

$$n = n_{\min} + n_{\text{span}} \times SF \times (TF \vee HF \vee LF) \tag{5}$$

$$TF = a \times T \times (T - T_{\min}) \times (T_{\max} - T), \quad T_{\min} < T < T_{\max} \tag{6}$$

$$HF = 0.00232 \times H^2 - 0.326 \times H + 11.4, \quad H > H_{\min} \tag{7}$$

$$LF = \begin{cases} \dfrac{8 \times 10^{-11} \times LW^4}{10^{-4} + \sqrt{LI^4}}, & \text{if } LF \leq 2 \\ 2, & \text{otherwise} \end{cases} \tag{8}$$

The environmental effects are mathematically represented in Eqs. (5) - (8), where TF denotes the influence of temperature, HF represents the effect of relative humidity, and LF captures the impact of light. These factors can assume values within the interval [0, 2]. Values below 1 restrict growth by directly limiting the concentration of the activator, while values above 1 promote growth. The parameters n_{min} and n_{span} are adjusted to align with the properties of the simulated mycelium, and only one environmental condition is examined at a time. Additionally, the substrate factor SF is introduced, to apply a small deviation of approximately $\pm 5\%$ in the matrix n, ensuring a non-uniform and biologically plausible substrate.

Furthermore, it has been observed that mycelium grows best in a particular temperature range representing a normal distribution (Omuse et al. 2022), while outside of this range, growth rates are slowed down by thermal stress (at high temperatures) or enzymatic inefficiency (at low temperatures). The optimal temperature values for optimal growth seem to be between 10 °C (T_{min}) and 35 °C (T_{max}) with a peak around 25 °C, as illustrated in Fig. 2a. In order to model this non-linear correlation between fungal growth and temperature, the Van Der Heide model has been used (Van der Heide et al. 2006), due to its simplicity and biological relevance in modeling the nonlinear relationship between temperature and fungal growth. This is described in Eq. (6), where TF represents the temperature's influence on mycelial growth rate due to a specific temperature (T), while α is a multiplier to limit TF on the specified range [0, 2].

Considering the humidity factor, it has been noted that the mycelium can grow vigorously as humidity rises. The reason is that the better hydration conditions allow nutrients to be transported and enzymatic reactions that are necessary for mycelial growth to occur. The Time of Wetness (TOW) model, which links mycelium growth rates to the duration of high relative humidity (e.g., above 80%), explains this phenomenon (Arya and Singh 2016). It has been observed that an area is not seen as prosperous and, hence, growth is hindered when the humidity level falls below 75%. In accordance with the findings from Arya and Singh (2016), which suggest a positive association of relative humidity to mycelial growth, the

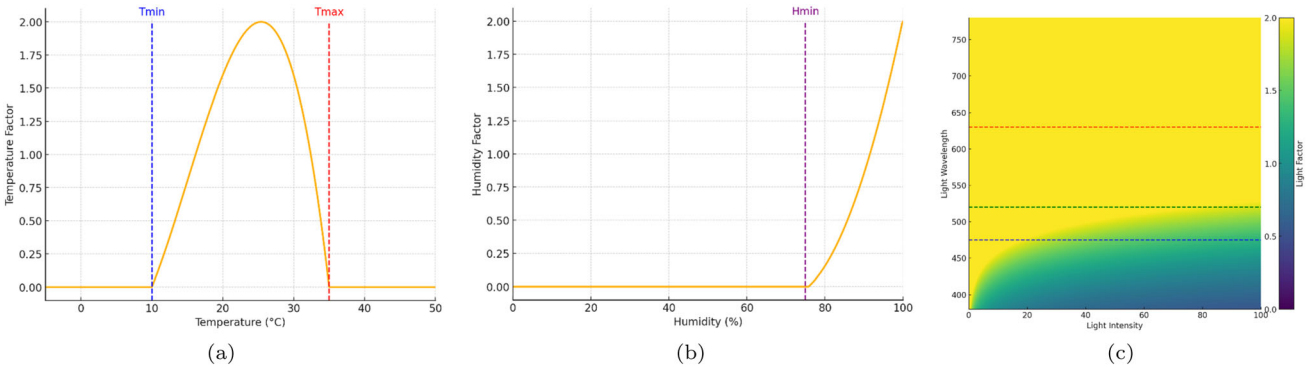


Fig. 2 Normalized effect of environmental factors integrated on the proposed model, namely (a) temperature, (b) humidity, and (c) light intensity and wavelength

mathematical model relating humidity and growth is presented in Eq. (7). Herein, H represents relative humidity in the span of [0, 100], while HF represents mycelial growth rate due to a specific humidity. The relationship between humidity and mycelial growth is depicted in Fig. 2b.

Light exerts a complex and wavelength-dependent influence on fungal mycelium growth. According to Lin et al. (2021), the growth and reproduction of Aspergillus oryzae were found to be highly sensitive to blue light (around 475 nm) under specific conditions, whereas green light (around 520 nm) and red light (around 630 nm) had minimal impact on the growth of mycelium. The inhibitory effects of light depend not only on the wavelength but also on its intensity, which is measured in $\mu mol\, photons\, m^{-2}s^{-1}$. For instance, blue light at intensities of 60 and 80 $\mu mol\, photons\, m^{-2}s^{-1}$ significantly suppressed mycelium growth, reducing colony diameters compared to darkness. In contrast, red and green light consistently showed negligible inhibition, with growth rates comparable to those observed in complete darkness.

This wavelength-intensity interplay reflects the selective activation of fungal photoreceptors, which are sensitive to specific light spectra. Blue light receptors, for example, are known to trigger inhibitory responses that reduce hyphal elongation and suppress conidium production. In the study, the number of conidia produced under blue light irradiation with and intensity of 80 $\mu mol\, photons\, m^{-2}s^{-1}$ was 57.4% lower than that in the darkness group. This stark reduction underscores the potential for blue light to regulate fungal reproduction in controlled environments. On the other hand, red light receptors appeared inactive under the tested conditions, aligning with the insensitivity observed in colony diameter and conidium yield. To capture this relationship, seen in Fig. 2c, the formula presented in Eq. (8) was employed, with LW to represent the light wavelength on the visible area [380, 780] and LI to represent the intensity on the range [0, 100].

2.3 Cellular automata implementation

Herein, the algorithm underlying the mycelium model is presented and thoroughly illustrated in Fig. 3 and described in Algorithm 1. To begin with, given that the concentrations of u and v must be determined prior to c, and the activator u is constrained to values within a specific spatial range from c, the first step involves defining the matrix Ω_c. This matrix is constructed by conditionally updating its values, specifically in cases where c exceeds 0.5, with a 5×5 inverse-square distance-weighted kernel (L_2), centered at the source (where $c > 0.5$) (line 3 of Algorithm 1). This simulates the propagation of the central "fired" cell's influence to its neighbors, with the effect diminishing as a function of distance from the source. Adjacent cells that are activated through this process are marked with $\Omega_c = 1$, enabling the activator u in these regions and facilitating its spread.

Algorithm 1 Mycelium model pseudo-code (iterative process)

1: $u, v, c, L_1, L_2 \leftarrow$ initialize
2: $n \leftarrow$ environmental conditions
3: **if** $(c > 0.5 \wedge L_2 > \Omega_c)$ **then** $\Omega_c = L_2$
4: $u_{new} = u + dt(N \circ (\text{conv}(u, L_1)) + \zeta(\kappa u + u^2 - \lambda uv)n)$
5: $v_{new} = v + dt(d(\text{conv}(v, L_1)) + \zeta(\mu u^3 - u))$
6: **if** $u_{new} \leq$ threshold **then**
7: $\quad a(u_{new}) = \xi$
8: **else**
9: $\quad a(u_{new}) = \xi - \theta(u_{new} - \text{threshold})$
10: **end if**
11: $c_{new} = c + dt(\zeta \nu c(a(u_{new}) - c)(c - 1))$
12: **if** $a(u_{old}) < 0$ **then** $c_{new} = c_{old} + q$
13: **if** $u_{new} < 0$ **then** $u_{new} = 0$
14: **if** $v_{new} < 0$ **then** $v_{new} = 0$
15: **if** $u_{new} > a_{max}$ **then** $u_{new} = a_{max}$
16: **if** $c_{new} > 1$ **then** $c_{new} = 1$
17: $v_{old} = v_{new}; \quad u_{old} = u_{new}; \quad c_{old} = c_{new}$

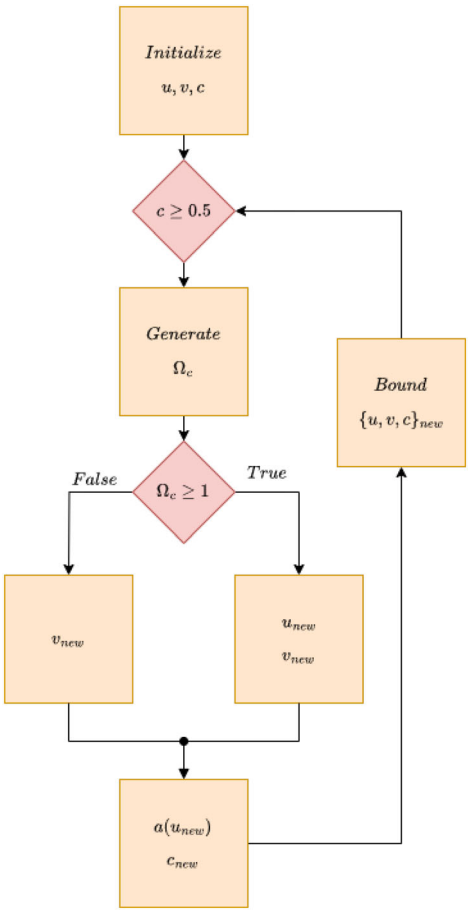


Fig. 3 Representation of the RD algorithm utilized in the proposed CA Mycelium Model. Begins with initialization of u, v, c. Following, Ω_c is generated based on the concentration of c. Should it be sufficient, apart from v_{new}, u_{new} is also generated promoting growth and $a(u_{new})$ is calculated. Finally, c_{new} is defined based on $a(u_{new})$ and each of the components $\{u, v, c\}_{new}$ are bounded to their corresponding ranges

After the variable space Ω_c is determined and moving on to the calculation of the activator and inhibitor, the Laplacian operator, which models diffusion across space (O'Reilly and Beck 2006), is applied to u and v, by convolving the system with a 3×3 Laplacian kernel (L_1) (lines 4 and 5). This approach allows for the parallel update of the concentrations of both the activator u and the inhibitor v, along with the calculation of the corresponding reaction terms of u and v. Also, in this state of the CA, the growth matrix n is integrated with the reaction terms of the activator u, allowing the model to simulate growth under distinct environmental influences. It is noteworthy that the calculation of the concentration of each RD component involves the time derivative, and consequently a relative time unit $dt = 10^{-5}$ was applied to both the reaction and diffusion terms.

Following this, the function $a(u)$ is defined. Depending on whether the activator's concentration meets the required threshold, set to 1, c is subsequently updated (lines 6-11). Finally, the concentrations of u, v, and c are evaluated to ensure they remain within their limits (lines 12-16). If any concentration falls outside the allowable range, it is bounded properly. In cases where the concentration c is negative, a scenario that, while biologically implausible, can arise due to the dynamics of $a(u)$ in discrete time and space, the concentration is not immediately clamped to 0 or 1. Instead, a small negative or positive noise term is introduced, respectively, to ensure continued growth under the constraints of discrete time and space. This approach maintains the system's stability while accommodating the inherent limitations of the modeling framework.

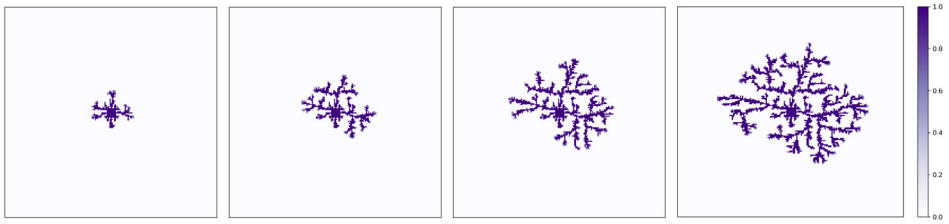
2.4 CA response under different conditions

The response of the employed CA model to different environmental conditions is evaluated in this Section. The precise calculations allow the simulation of mycelial behavior under varied conditions, making the model a robust tool for predictive analysis and optimization.

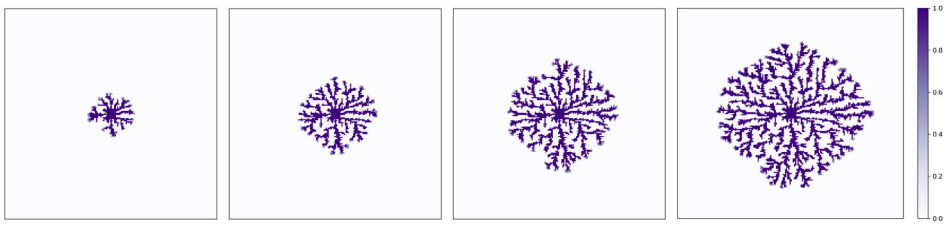
In Fig. 4a-b, the influence of varying temperatures on mycelium growth is illustrated. Optimal growth is observed at $T = 25°C$, while minor deviations from this temperature, such as $T = 29°C$, do not significantly impede growth and can still support favorable development. In contrast, substantial deviations lead to reduced growth rates, primarily due to enzymatic inefficiency or thermal stress, as evidenced in the case of $T = 15°C$.

Following, Fig. 4c-d present the mycelium growth under different relative humidity levels. A very high relative humidity ensures mycelium development, like in the case of $H = 97\%$, where a proper growth can be observed. On the other hand, insufficient humidity can inhibit growth, as observed in the case of $H = 84\%$.

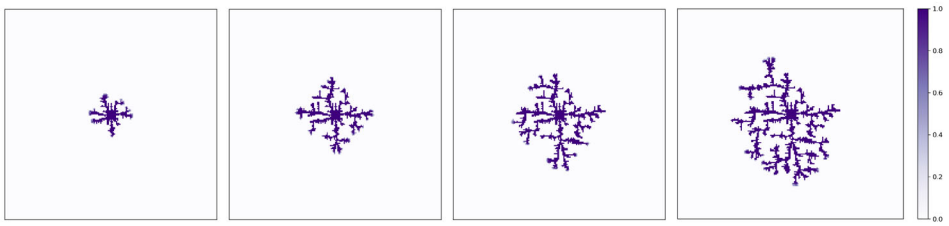
Finally, the influence of light on mycelium growth is depicted in Fig. 4e-f. Light affects mycelial development primarily through its wavelength (LW) and intensity (LI). Short wavelengths, such as blue light, particularly at high intensity, are demonstrated to inhibit growth ($480\,nm$ at an intensity of $80\ \mu mol\,photons\,m^{-2}s^{-1}$), whereas red light, especially at low intensity, exhibits minimal inhibitory effects ($630\,nm$ at an intensity of 0). This is clearly demonstrated by the limited growth observed under blue light compared to the red light scenario.



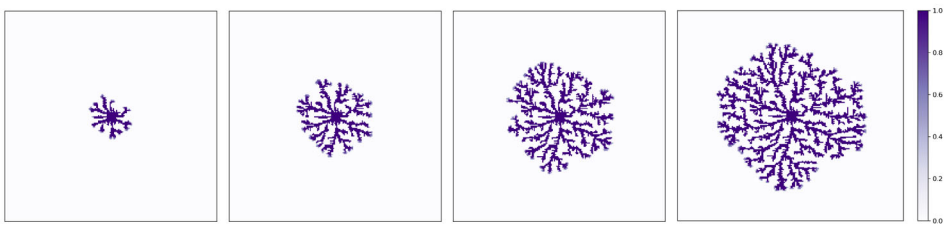
(a) Temperature's impact for $T = 15°C$



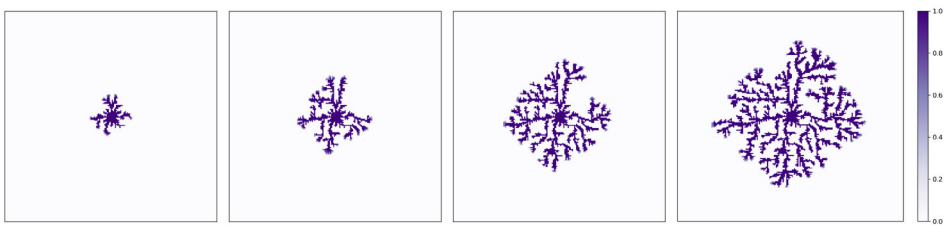
(b) Temperature's impact for $T = 29°C$



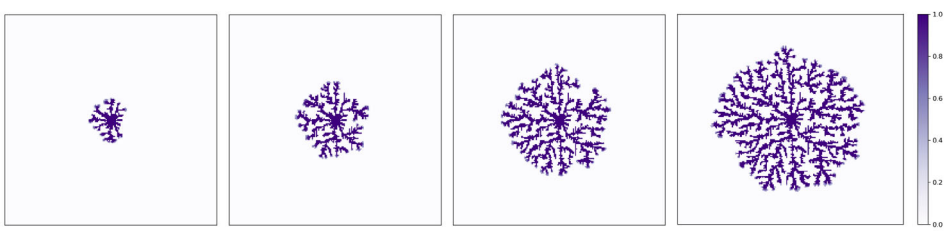
(c) Humidity's impact for $H = 84\,\%$



(d) Humidity's impact for $H = 97\,\%$



(e) Light's impact for $LW = 480\,nm$ and $LI = 80\,\mu mol\ photons\ m^{-2}s^{-1}$



(f) Light's impact for $LW = 630\,nm$ and $LI = 0\,\mu mol\ photons\ m^{-2}s^{-1}$

Fig. 4 Impact of diverse environmental conditions on mycelium growth every 15,000 timesteps, in a 200×200 lattice

3 Evaluating model as a reservoir network

Modeling itself can reproduce plausible results. However, it is difficult to accurately simulate mycelium growth because of the complex and dynamic interactions between hyphae, the fundamental structural units of mycelium (namely the tips), and their surroundings. In most fungi, hyphae serve as the primary mechanism for vegetative growth and are collectively referred to as mycelium. Being able to identify which parameters can be adjusted accordingly to match realistic fungi behavior can lead to an accurate reservoir network with proper computational properties based on its small-world properties.

3.1 Hyphae information extraction

Given the model's emphasis is on generalization, its capacity to represent a variety of mycelium structures, greater attention should be directed towards the parameters d, κ, λ, and μ, which critically influence the topological characteristics of the hyphae, particularly the count, width, and length of the hyphal tips and ultimately serve as tuning parameters. To quantify the specified characteristics, an algorithm has been developed to generate the necessary metrics for adjusting the model to the corresponding real mycelium as it calculates the discussed hyphal features of an experimental mycelium image. In such a way, it is possible to infer the fungal network's efficiency, flexibility, and general health under different circumstances, leading to modeling simulations that closely resemble actual observations of fungal growth.

Initially, the mycelium core must be determined, as it is the central hub from which the hyphal strands occur. This process is achieved by applying a convolutional kernel to detect regions within the matrix c, which represents the mycelium concentration. This process identifies areas characterized by eight high-concentration neighbors that signify the core of the structure. Subsequently, a specified concentration threshold is applied to these regions to validate the true core and not other densely connected regions. This threshold is dynamically determined based on the concentration range, the predefined number of growth iterations in the model, and the initial conditions, which influence the rate at which the maximum concentration within the range is achieved over the specified iterations.

In order to detect the first characteristic (the hyphal count), the algorithm must locate the hyphal tips by employing neighbor count analysis. They are represented as the array cells of c with a limited number of neighbors (three or fewer) having less than a specific concentration. This concentration is relative to the concentration of c near its edges. To compute the width of each tip, which reflects the robustness and vitality of fungal growth, a kernel based on the inverse square of the distance from the core is convolved with the tip locations of c. This convolution measures the intensity distribution around each tip, serving as a proxy for width. Finally, to determine the length of each tip that gives information about the magnitude and direction of fungal growth, a breadth-first search (BFS) algorithm is utilized. Starting from each identified tip, the BFS investigates the shortest route from the tip to the mycelium core, tracing the hyphae backwards to the core. The search terminates when a core cell is detected, and the path length is then measured to indicate how far the hyphal tip extends from the core. In Fig. 5 the metrics are visualized on a 2D plane, taking into account the default values for the tuning parameters. In Fig. 5b the mycelium core has been identified, while in Fig. 5c-d each cell represents the tip width and tip length, respectively.

The visualized metrics in Fig. 6 illustrate how variations in the four key parameters (d, κ, λ, and μ) by $\pm 5\%$ affect hyphal growth characteristics, including tip count, average width, and average length. These parameters align with their roles in Eqs. (1) - (4), demonstrating their influence on hyphal morphology. The parameter d, representing the ratio between the inhibitor's and activator's diffusion rates, enhances inhibitor diffusion, allowing it to spread more extensively across the grid. This forces the activator into narrower pathways, leading to increased branching and tip count while reducing hyphal width. The parameter κ amplifies activator concentration, promoting both wider and longer hyphal tips. In contrast, λ reduces width and length by lowering activator concentration while simultaneously decreasing tip count by diminishing the inhibitor's influence in the reaction terms. Lastly, μ suppresses all three characteristics, tip width, length, and count, by directly increasing inhibitor concentration, thereby limiting overall growth and structural expansion.

3.2 Small-world properties

With the mycelium model and feature extraction established, this work explores how the unique characteristics of mycelium's structure can serve as a basis for unconventional computing that is resource-, cost-, and power-efficient. Its intricate structure, which is marked with high and random connectivity, showcases a small-world topology, mainly high connectivity at its core sparsity connections near its edges, which is a useful attribute for building reservoir networks, as it has been shown to improve the echo-state property (Kawai et al. 2019). Additionally, it has been shown that the echo-state property in reservoir computing is improved by small-world characteristics, which are characterized by high node clustering and short

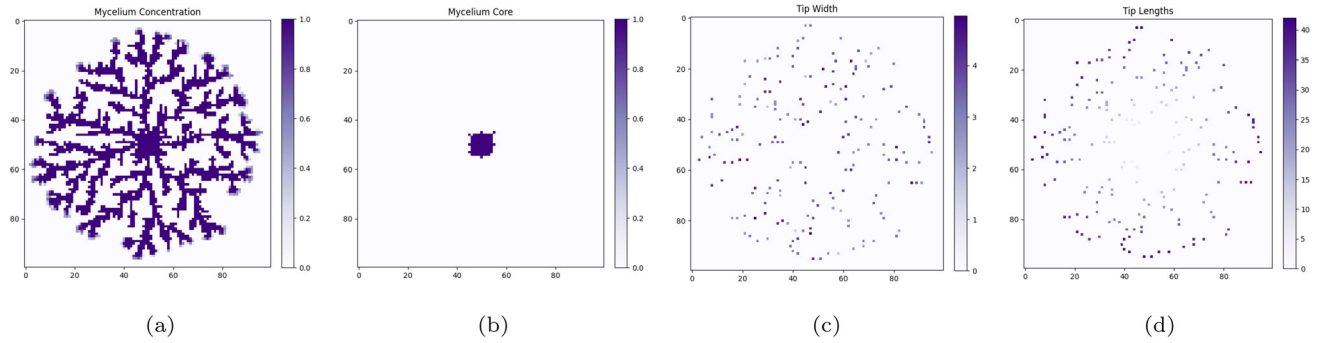


Fig. 5 Basic mycelium characteristics produced by the hyphae extraction algorithm, namely: **a** concentration, **b** core, **c** tip width, and **d** tip length

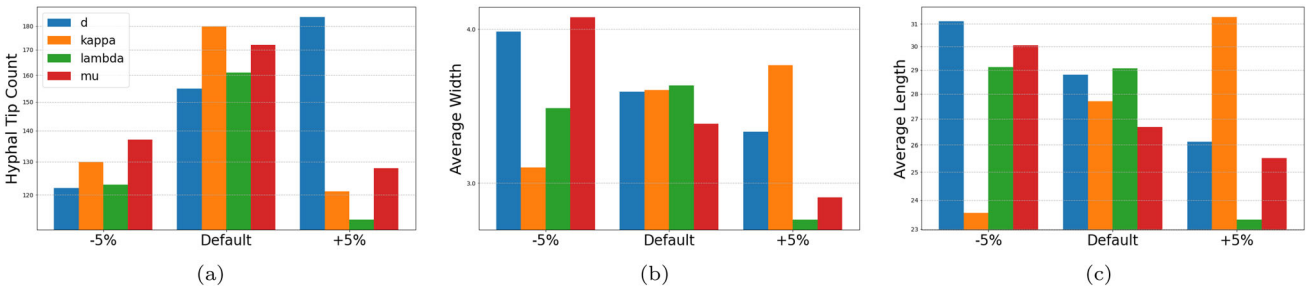


Fig. 6 Change in: **a** hyphal tip count, **b** average tip width and **c** average length (*log* scale) by adjusting the model's tuning parameters to $\pm 5\,\%$ of their default values, being $d = 29$, $\kappa = 0.8$, $\lambda = 0.8$ and $\mu = 3.5$

path lengths between nodes (Tsakalos et al. 2021; Manneschi et al. 2021; Bassett and Bullmore 2017).

In order to evaluate the mycelium model's feasibility as a reservoir, the average path length (*APL*) and average clustering coefficient (C_i), two important metrics representative of small-world characteristics, were computed. The results were shown for 30 different runs, as presented in Table 1, to ensure fidelity and account for the small randomness inherent in the model. First, an adjacency matrix A, derived from the mycelium's concentration matrix c, has been created in order to obtain these metrics. In addition, the model was treated as an unweighted graph, with each cell of the matrix c having a significant concentration (greater than 0.5) being considered as a node.

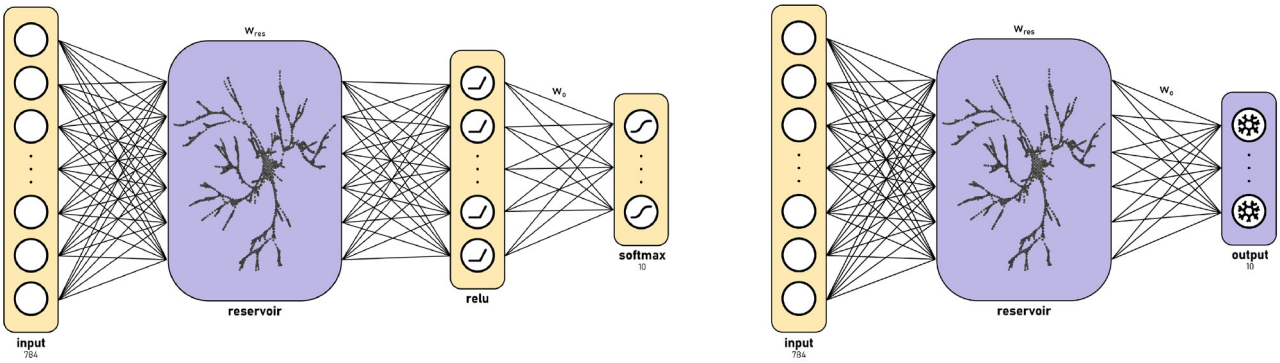
The *APL* (Average Path Length) is a critical metric for evaluating the efficiency of information propagation within a network. It is defined as the mean of the shortest paths between all pairs of nodes and is computed using Eq. (9), where $d(i, j)$ represents the shortest path distance between nodes i and j, and N denotes the total number of nodes. A low *APL* (greater than but close to 1) signifies rapid information transmission across the network, a fundamental characteristic for enabling the responsive dynamics required in reservoir computing (Nakajima and Fischer 2021).

Table 1 Small-World characteristics for 30 runs of the model

Average path length		Clustering coefficient	
1–1.65	16–1.72	1–0.19	16–0.21
2–1.58	17–1.69	2–0.18	17–0.20
3–1.71	18–1.63	3–0.20	18–0.19
4–1.67	19–1.74	4–0.21	19–0.18
5–1.82	20–1.68	5–0.19	20–0.22
6–1.73	21–1.77	6–0.20	21–0.21
7–1.79	22–1.75	7–0.18	22–0.19
8–1.66	23–1.71	8–0.22	23–0.20
9–1.74	24–1.69	9–0.19	24–0.18
10–1.81	25–1.72	10–0.21	25–0.20
11–1.83	26–1.70	11–0.20	26–0.19
12–1.68	27–1.76	12–0.18	27–0.21
13–1.85	28–1.73	13–0.19	28–0.20
14–1.72	29–1.78	14–0.21	29–0.18
15–1.77	30–1.74	15–0.20	30–0.19
Avg - 1.729		**Avg - 0.198**	

$$APL = \frac{1}{N(N-1)} \sum_{i \neq j} d(i,j) \tag{9}$$

The clustering coefficient quantifies the propensity of nodes to form densely interconnected groups, facilitating



(a) Contemporary (b) Biological

Fig. 7 The two Reservoir network architectures employed, differing in their output

localized and intricate dynamics within the network. For a given node i, C_i is determined by Eq. (10), where $\deg(i)$ represents the degree of node i. A high clustering coefficient (less than but close to 1) is a hallmark of small-world networks and enhances the network's capacity to generate diverse, context-dependent responses, which are crucial for applications in reservoir computing (Gallicchio et al. 2018).

$$C_i = \frac{\sum_{j,k} A_{ij}A_{jk}A_{ki}}{\deg(i)(\deg(i)-1)} \tag{10}$$

The *APL* values demonstrate an average of 1.729 across all configurations, with a range extending from 1.58 to 1.85, while the C_i values exhibit an average of 0.198, ranging from 0.18 to 0.22. To assess these metrics and determine whether the resulting topology exhibits small-world characteristics, the small-world coefficient must be computed. If this coefficient, derived from Eq. (11), exceeds one ($S > 1$), results in the topology being classified as small-world (Hung and Wang 2010).

$$S = \frac{C_{\text{graph}}}{C_{\text{random}}} \cdot \frac{L_{\text{random}}}{L_{\text{graph}}} \tag{11}$$

Herein, L_{graph} and C_{graph} represent the average path length and average clustering coefficient of the graph, respectively, while L_{random} and C_{random} denote the corresponding metrics for a random graph with an equivalent number of nodes and edges. These metrics were computed using the NetworkX Python package (Hagberg et al. 2008), which supported the efficient generation of a sufficiently large sample (up to 30 instances) of random graph metrics. The results not only yielded a small-world coefficient significantly greater than one but also revealed values consistently ranging between 30 and 1000, highlighting the mycelium network's strong potential as a small-world topology. These results indicate the network's capacity for rapid data propagation and structural adaptability. The high clustering coefficient and optimal average path length confirm its suitability for dynamic, nonlinear applications.

4 Reservoir computing with mycelium CA-based networks

The biological mycelium has been the subject of research regarding its electrical activity (Adamatzky 2022, 2018b), which has demonstrated its ability to propagate information and electrical signals across its network. In particular, studies have revealed that mycelial networks are capable of performing basic computational operations, such as implementing Boolean logic gates (Adamatzky et al. 2022). When combined with its inherent small-world properties, marked by high clustering coefficients and short path lengths, these properties suggest mycelium could be an effective substrate for reservoir computing.

To evaluate this potential, two different mycelium-based reservoir networks, illustrated in Fig. 7, have been defined and their performance was evaluated on the classification of the MNIST dataset, thereby constituting a comprehensive testbed. Both networks start with a fully connected input layer, followed by a mycelium-based reservoir layer. The construction of the reservoir layer utilizes the adjacency matrix A analyzed in Section 2 as its weight matrix. The response of the reservoir layer to each image, for a time window of $6\,ms$, is captured and, along with the corresponding labels, is used to train the output weights (W_O) of the networks.

4.1 Reservoir layer neuron

The electrical activity observed in mycelium exhibits striking similarities to biological neuronal spiking behavior, characterized by distinct phases of depolarization, repolarization, and a refractory period (Adamatzky 2018a). Thus, a biological neuron model is selected in the reservoir

layer's nodes to properly replicate key neuronal phenomena, such as action potentials and threshold behavior. There are various such models, including the Leaky Integrate-and-Fire (LIF), the Izhikevich (IZH), and the FitzHugh-Nagumo (FHN); yet, this work is mostly focused on the FHN model, as prior studies have demonstrated its effectiveness in mimicking the electrical activity of mycelium (Adamatzky et al. 2022).

The model in question is an efficient representation of neuronal dynamics, capturing the essential features of spiking behavior while remaining computationally inexpensive. The FHN model is a two-dimensional simplification of the Hodgkin-Huxley equations (Phillipson and Schuster 2005), describing the activation and inactivation of ion channels in neuronal membranes. It consists of two coupled differential equations, showcased in Eqs. (12) - (15) (He et al. 2020). By incorporating it into the network, this work aims to better emulate the electrical properties of biological neurons, thereby providing a more realistic framework for evaluating the computational capabilities of mycelium-based reservoirs.

$$\frac{dv(t)}{dt} = v(t) - \frac{1}{3}v(t)^3 + w(t) + I,\ v(t) < v_{\text{thresh}} \tag{12}$$

$$\frac{dw(t)}{dt} = \epsilon(a - v(t) - bw(t)),\ v(t) < v_{\text{thresh}} \tag{13}$$

$$v = v_{\text{reset}}, \quad v(t) \geq v_{\text{thresh}}. \tag{14}$$

$$w = w(t) + d, \quad v(t) \geq v_{\text{thresh}}. \tag{15}$$

$$I = g_{ext} \times in_i + g_{local} \times sum(W_{fired}) \tag{16}$$

The FHN equations describe the dynamics of the membrane potential $v(t)$ and the recovery variable $w(t)$, capturing the essential features of neural activity. Equation (12) governs the evolution of $v(t)$, where the term $v(t) - \frac{1}{3}v(t)^3$ models the rapid activation and inactivation of sodium ion (Na^+) channels, while $w(t)$ represents the slower recovery process mediated by potassium ion (K^+) channels. The external current I is a key input to the system, representing the total synaptic input or external stimulation received by the neuron. This current directly influences the membrane potential, driving the neuron toward firing or resting states. Equation (13) describes the recovery variable $w(t)$, with ϵ being a small parameter that ensures the recovery process occurs more slowly than changes in $v(t)$. The parameters a and b determine the resting state of the neuron and the sensitivity of the recovery process to changes in $v(t)$, respectively.

When $v(t)$ exceeds the threshold v_{thresh}, the neuron emits a spike, and the system resets according to Eqs. (14) and (15). At this point, we can deduce that a neuron has "fired", then $v(t)$ is set to v_{reset}, and $w(t)$ is increased by a constant d to account for the refractory period following a spike. As a result, the neuron outputs I as a response to its inputs, and the FHN model captures the interplay between membrane potential dynamics, recovery processes, and external stimuli, providing a comprehensive description of neural spiking behavior.

To enhance the realism of the spiking behavior of the nodes, the Nelder-Mead method (Nelder and Mead 1965) was applied to adjust the parameters ϵ, a, and b in the FHN equation set. This optimization process aimed to align the model's output with specific spiking characteristics, including period, amplitude, duration, depolarization rate, repolarization rate, and refractory period, as observed in studies of fungal electrical activity (Adamatzky 2018a). The optimization was guided by minimizing the mean squared error (MSE) between the simulated FHN dynamics and the target spiking features, ensuring a close match to the empirical data. By iteratively refining the parameters, the algorithm successfully captured the essential electrical characteristic of fungal spiking, as shown in Fig. 8, and more specifically the spiking characteristics of the oyster fungi Pleurotus djamor.

Building upon the network implementation, the input traversing through the reservoir layer influences the firing dynamics of the FHN neurons, with a key conceptual note being that the input connected to the reservoir nodes is incorporated as part of the external current I in the FHN model, showcased in Eq. (12). Specifically, the total input current I for each FHN neuron is calculated as the sum of two components, as described in Eq. 16: the external input and the synaptic input.

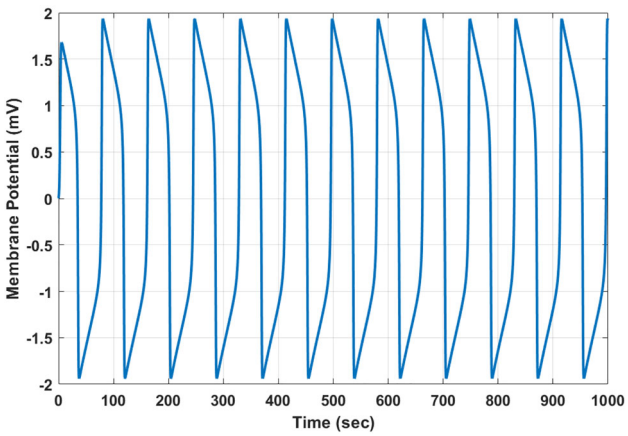
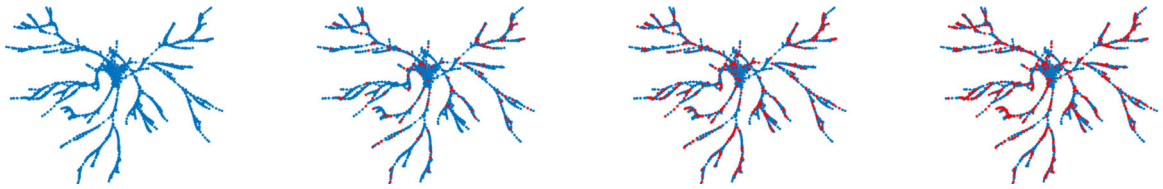


Fig. 8 Spiking activity of FitzHugh-Nagumo neurons with ϵ=0.0335, a=0.1034, b=1.1887, and I_{ext}=0.0645, v_{thresh}=1.3 mV and d=0.01 featuring: Period (*sec*) 82.8172, Amplitude (*mV*) 1.9170, Duration (*sec*) 82.8172, Depolarization Rate (*mV/sec*) 0.0295, Repolarization Rate (*mV/sec*) 0.0294, and Refractory Period (*sec*) 83.5223, resembling fungal spiking activity under thermal stimuli in Adamatzky (2018a)



(a) 0 ms (b) 2 ms (c) 4 ms (d) 6 ms

Fig. 9 Spiking activity of the reservoir layer with FitzHugh-Nagumo nodes, for a random sample of the MNIST dataset (number 4), observed in a time window of 6 ms. Reservoir's response: **a** 0 spikes, **b** 270 spikes, **c** 617 spikes, and **d** 882 spikes

The external input represents the contribution from the input layer, scaled by a factor g_{ext}, and corresponds to images or other stimuli, denoted as in, where i indexes the current input. This external input is directly injected into the neurons as part of the driving current. The synaptic input, on the other hand, represents the input received from neighboring neurons within the reservoir layer that have fired, calculated as local_gain $\times$ sum(W_{fired}). The parameter g_{local} scales this synaptic input, allowing control over its influence on the neuron's dynamics.

Together, these components define the total current I that drives the FHN neurons, integrating both external stimuli and internal network dynamics. The temporal evolution of this current over a 6 ms window is illustrated in Fig. 9, using a random sample from the dataset. The figure demonstrates the influence of the external current in the initial phase of the response (around 2 ms), where it stimulates neurons at random positions, generating a total of 270 spikes. As the response progresses, the activity propagates to neighboring neurons, particularly in the later phase of the time window (between 4 ms and 6 ms). During this period, the spiking activity increases significantly, reaching a peak of 882 spikes. This pattern highlights the interplay between external stimulation and emergent network dynamics, showcasing how localized input can lead to widespread neural activation over time.

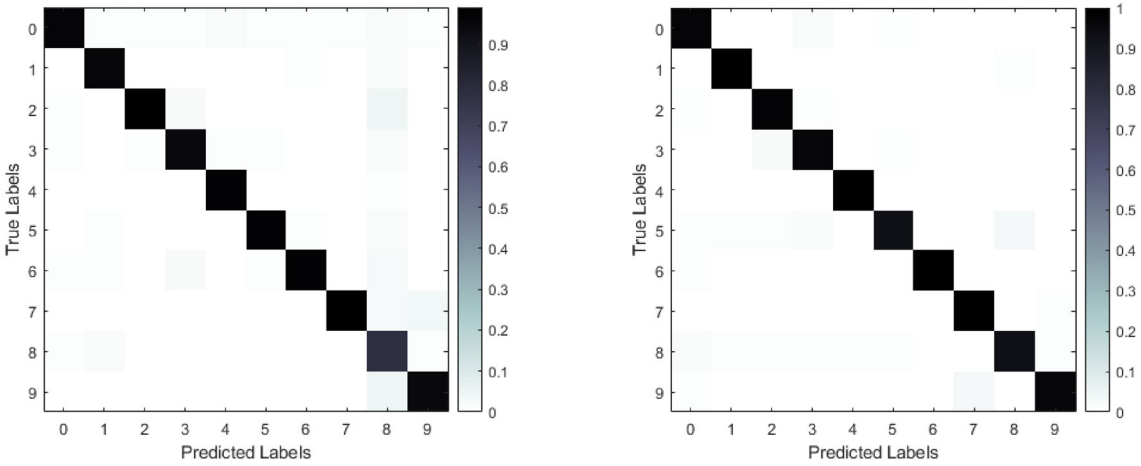
4.2 Output layers

Regarding the output layers, two options have been employed: a purely biologically-inspired approach, followed by a contemporary one. The fully biologically-inspired approach incorporates changes in both its nodes and training method. This network consists of a reservoir layer and an output layer, both composed of FHN neurons, with the output layer containing 10 neurons (Fig. 7b). Training is based on the precise spike timing, a key aspect of biological neural communication. During training, the spike timings of both the reservoir and output layers are recorded. Neurons in the output layer that exhibit synchronous firing with those in the reservoir layer are reinforced, while those firing asynchronously are penalized. This mechanism is governed by the ReSuMe (Remote Supervised Method) learning rule (Kasiński and Ponulak 2006), which is mathematically expressed in Eq. (17).

$$\frac{dw(t)}{dt} = (S_d(t) - S_o(t))[\alpha + \int_0^\infty W(s)S_{in}(t-s)\,ds] \quad (17)$$

The equation describes the rate of change of synaptic weights $\frac{dw(t)}{dt}$ over time, driven by the difference between the desired spike output $S_d(t)$ and the actual spike output $S_o(t)$. The learning process is further modulated by a constant learning rate α and an integral term $\int_0^\infty W(s)S_{in}(t-s)\,ds$, which accounts for the influence of past input spikes S_{in} weighted by a kernel function $W(s)$. By rewarding synchronized spike timing and penalizing otherwise, ReSuMe learning aligns with the principles of spike-timing-dependent plasticity (STDP) (Ponulak and Kasiński 2010; Passias et al. 2024), enabling the network to adapt its synaptic weights in a biologically plausible manner. This approach not only enhances the network's ability to process temporal information but also bridges the gap between artificial and biological neural systems, emphasizing the importance of precise spike timing in learning and memory.

The second approach includes a fully connected ReLU layer, followed by a fully connected softmax layer at the output (Fig. 7a), which produces probabilities indicating the likelihood of an image belonging to each of the ten classes of the dataset. Regarding the training of this network, a standard cross-entropy loss function was employed to evaluate the performance of the classification task. This loss function directly compares the predicted output probabilities to the desired one-hot encoded labels, quantifying the discrepancy between the model's predictions and the ground truth. To optimize the parameters of this network, the Adam optimizer (Zhang 2018) has been utilized, a widely adopted optimization algorithm based on momentum-based gradient descent known for its efficiency and robustness in the training of deep learning (DL) models.



(a) Biological (b) Contemporary

Fig. 10 Confusion matrices for all the samples of the MNIST dataset for different network architectures. Results are column-normalized and scaled to [0, 1]

Table 2 Results of different reservoir nodes employed, in both the biological and contemporary architectures

Output architecture & training method	Reservoir node	Accuracy (%)
Biological	LIF	91.84
	FHN	90.51
	IZH	92.32
Contemporary	LIF	96.9
	FHN	96.99
	IZH	97.09

4.3 Results comparison

Moving on to the results, each network was evaluated having a 70/30 training/testing sample ratio, randomly distributed, for 50 epochs. The results demonstrate the effectiveness of mycelium-based reservoir computing for classifying the MNIST dataset, with all network configurations achieving high accuracies, as showcased by the confusion matrices depicted in Fig. 10. The pure biological configuration, which combined the FHN model with the ReSuMe learning rule for spike-timing-dependent plasticity, achieved an accuracy of 90.51%, whereas the contemporary architecture achieved 96.99%. The results demonstrate the potential of leveraging mycelium growth for cognitive tasks, achieving high accuracy in MNIST classification. Furthermore, it is apparent that integrating deep learning (DL) with an additional ReLU layer, coupled with a gradient descent-based training algorithm, outperforms a purely biological approach. This highlights the significance of combining biologically inspired architectures with advanced, optimized training methodologies. This observation is also supported by the comparative analysis with other biological neuron models, presented in Table 2, namely LIF and Izhikevich, employed in reservoir and output layers. Herein, where we can see similar results for each configuration respectively with the contemporary architecture reaching up to 97.09% accuracy, utilizing the IZH neuron model, closely followed by the FHN and LIF models.

5 Conclusions

In this work, a comprehensive framework for modeling mycelium growth and exploring its computational capabilities has been presented. It is based on the Cellular Automata (CA) theory enhanced with reaction-diffusion (RD) processes in order to vigorously capture the intricate growth dynamics of mycelium under various environmental factors such as temperature, humidity, and light, useful for understanding its behavior in real-world conditions. On top of that, a hyphae information algorithm has been included in the framework to extract key features useful for appropriate tuning of the model to real structures. Taking into account the result that the mycelium model has generated with all the aforementioned features, it has been proven that the structure is capable of serving as a reservoir layer for unconventional computing, leveraging its electrical activity. More specifically, a test case of classification of the MNIST dataset has been conducted with accuracy up to 97.09%. To enhance the biological fidelity of the designed network and closely mimic the electrical activity observed in real mycelium, the Fitz-Hugh-Nagumo (FHN) neuron model was integrated into the reservoir layer. In this direction, the gap between

biological systems and artificial intelligence is being further bridged, opening new avenues for the development of bio-inspired computing systems utilizing sustainable Engineered Living Materials (ELMs) for future computing applications.

We aim to improve the model by including data from different fungal species and studying how they respond to environmental changes. Additionally, the exploration of other machine learning tasks and datasets will help to fully investigate the capabilities of mycelium-based computing systems. The next step is implementing the reservoir layer in novel hardware for real-time, low-power computation and developing a digital twin of real mycelial structures replicated from the proposed model.

Author contributions All authors contributed equally to the study conception and design, the material preparation, data collection and analysis, and the manuscript writing and editing. All authors read and approved the final manuscript.

Funding Open access funding provided by HEAL-Link Greece. This work has been supported by the framework of the FUNGATERIA project, which has received funding from the European Union's HORIZON-EIC-2021-PATHFINDER CHALLENGES program under grant agreement No. 101071145.

Availability of data and materials The datasets generated during and/or analyzed during the current study are available from the corresponding author on reasonable request.

Declarations

Conflict of interest The authors have no Conflict of interest to declare that are relevant to the content of this article.

Ethics approval Not applicable.

Open Access This article is licensed under a Creative Commons Attribution 4.0 International License, which permits use, sharing, adaptation, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if changes were made. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit http://creativecommons.org/licenses/by/4.0/.

References

Adamatzky A (2018a) On spiking behaviour of oyster fungi pleurotus djamor. Sci Rep 8(1):7873

Adamatzky A (2018b) Towards fungal computer. Interface Focus 8(6):20180029

Adamatzky A (2022) Language of fungi derived from their electrical spiking activity. R Soc Open Sci 9(4):211926

Adamatzky A (2023) Fungal machines: sensing and computing with fungi, vol 47. Springer

Adamatzky A, Ayres P, Belotti G, et al (2019) Fungal architecture. arXiv preprint arXiv:1912.13262

Adamatzky A, Ayres P, Beasley AE et al (2022) Logics in fungal mycelium networks. Log Univers 16(4):655–669

Aiduang W, Jinanukul P, Thamjaree W et al (2024) A comprehensive review on studying and developing guidelines to standardize the inspection of properties and production methods for mycelium-bound composites in bio-based building material applications. Biomimetics 9(9):549

Angelova GV, Brazkova MS, Krastanov AI (2021) Renewable mycelium based composite-sustainable approach for lignocellulose waste recovery and alternative to synthetic materials-a review. Z Naturforschung C 76(11–12):431–442

Arya G, Singh J (2016) The mathematical isopleth model for fungi growth in building material. J Drug Deliv Ther 6(4):21–26

Ayres P, Rigobello A, You-Wen J, et al (2022) Investigating a design and construction approach for fungal architectures. In: Design modelling symposium Berlin, Springer, pp 571–583

Bassett DS, Bullmore ET (2017) Small-world brain networks revisited. Neuroscientist 23(5):499–516

Beiler KJ, Durall DM, Simard SW et al (2010) Architecture of the wood-wide web: Rhizopogon spp. genets link multiple douglas-fir cohorts. New Phytol 185(2):543–553

Boswell GP (2008) Modelling mycelial networks in structured environments. Mycol Res 112(9):1015–1025

Carver I, Boswell GP (2008) A lattice-free model of translocation-induced outgrowth in fungal mycelia. IAENG Int J Appl Math 38(4):173–179

Chatzinikolaou TP, Fyrigos IA, Ntinas V, et al (2022) Wave cellular automata for computing applications. In: 2022 IEEE international symposium on circuits and systems (ISCAS), pp 3463–3467, https://doi.org/10.1109/ISCAS48785.2022.9937915

Chatzinikolaou TP, Karamani RE, Fyrigos IA et al (2024) Handling sudoku puzzles with irregular learning cellular automata. Nat Comput 23(1):41–60. https://doi.org/10.1007/s11047-024-09975-4

Chatzipaschalis IK, Chatzinikolaou TP, Fyrigos IA et al (2023) Memristor-based cellular automata for natural language processing. In: 2023 30th IEEE international conference on electronics. IEEE, Circuits and Systems (ICECS), pp 1–4

Chatzipaschalis IK, Tompris I, Rallis K, et al (2024) Mycelium-based elm digital twin implemented in fpga. In: International conference on cellular automata for research and industry, Springer, pp 265–276

Codd EF (2014) Cellular automata. Academic press

Dale M, O'Keefe S, Sebald A et al (2021) Reservoir computing quality: connectivity and topology. Nat Comput 20:205–216

Dehshibi MM, Adamatzky A (2021) Electrical activity of fungi: spikes detection and complexity analysis. Biosystems 203:104373

Elsacker E, Zhang M, Dade-Robertson M (2023) Fungal engineered living materials: the viability of pure mycelium materials with self-healing functionalities. Adv Func Mater 33(29):2301875

Fricker MD, Heaton LL, Jones NS, et al (2017) The mycelium as a network. The fungal kingdom pp 335–367

Gallicchio C, Micheli A, Pedrelli L (2018) Design of deep echo state networks. Neural Netw 108:33–47

Hagberg A, Swart PJ, Schult DA (2008) Exploring network structure, dynamics, and function using networkx. Tech. rep., Los Alamos National Laboratory (LANL), Los Alamos, NM (United States)

He Z, Li C, Chen L et al (2020) Dynamic behaviors of the fitzhugh-nagumo neuron model with state-dependent impulsive effects. Neural Netw 121:497–511

Van der Heide T, Roijackers RM, Van Nes EH et al (2006) A simple equation for describing the temperature dependent growth of free-floating macrophytes. Aquat Bot 84(2):171–175

Hung SW, Wang AP (2010) Examining the small world phenomenon in the patent citation network: a case study of the radio frequency identification (rfid) network. Scientometrics 82(1):121–134

Jesse H (2023) Understanding the best humidity for mycelium growth. https://amhuru.com/understanding-the-best-humidity-for-mycelium-growth/

Karamani RE, Fyrigos IA, Tsakalos KA et al (2021) Memristive learning cellular automata for edge detection. Chaos, Solitons Fractals 145:110700

Karana E, Blauwhoff D, Hultink EJ, et al (2018) When the material grows: a case study on designing (with) mycelium-based materials. Int J Des 12(2)

Kasiński A, Ponulak F (2006) Comparison of supervised learning methods for spike time coding in spiking neural networks. Int J Appl Math Comput Sci 16(1):101–113

Kawai Y, Park J, Asada M (2019) A small-world topology enhances the echo state property and signal propagation in reservoir computing. Neural Netw 112:15–23

Lin S, Qin H, Zhang X et al (2021) Inhibition of aspergillus oryzae mycelium growth and conidium production by irradiation with light at different wavelengths and intensities. Microbiol Spectr 9(1):10–1128

Maass W (2011) Liquid state machines: motivation, theory, and applications. Computability in context: computation and logic in the real world pp 275–296

Manneschi L, Lin AC, Vasilaki E (2021) Sparce: improved learning of reservoir computing systems through sparse representations. IEEE Trans Neural Netw Learn Syst 34(2):824–838

Molitorisová A, Monaco A (2023) Innovating food law with mycelium: Eu regulations. SSRN

Molitorisová A, Monaco A, Purnhagen KP (2021) An analysis of the regulatory framework applicable to products obtained from mushroom and mycelium. Available at SSRN 3955899

Mora-Boza A, Acosta S, Puertas-Bartolomé M (2023) Biopolymers for the development of living materials for biomedical applications. In: Biopolymers. Elsevier, p 263–294

Nakajima K, Fischer I (2021) Reservoir computing. Springer

Nelder JA, Mead R (1965) A simplex method for function minimization. Comput J 7(4):308–313

Ntinas V, Karamani RE, Fyrigos IA, et al (2020) Cellular automata coupled with memristor devices: a fine unconventional computing paradigm. In: 2020 international conference on electronics, information, and communication (ICEIC), IEEE, pp 1–4

Omuse ER, Niassy S, Wagacha JM et al (2022) Suitable models to describe the effect of temperature on conidial germination and mycelial growth of metarhizium anisopliae and beauveria bassiana. Biocontrol Sci Tech 32(3):281–298

O'Reilly RC, Beck JM (2006) A family of large-stencil discrete laplacian approximations in three-dimensions. Int J Numer Methods Eng pp 1–16

Passias A, Tsakalos KA, Kansizoglou I et al (2024) A biologically inspired movement recognition system with spiking neural networks for ambient assisted living applications. Biomimetics 9(5):296

Pavlidis N, Perifanis V, Chatzinikolaou TP, et al (2023) Intelligent client selection for federated learning using cellular automata. In: 2023 18th international workshop on cellular nanoscale networks and their applications (CNNA), pp 1–4, https://doi.org/10.1109/CNNA60945.2023.10652769

Phillipson PE, Schuster P (2005) A comparative study of the hodgkin-huxley and fitzhugh-nagumo models of neuron pulse propagation. Int J Bifurc Chaos 15(12):3851–3866

Ponulak F, Kasiński A (2010) Supervised learning in spiking neural networks with resume: sequence learning, classification, and spike shifting. Neural Comput 22(2):467–510

Roberts N, Adamatzky A (2023) Mining logical circuits in fungi. In: Fungal machines: sensing and computing with fungi. Springer, p 311–321

Sugimura K, Shimono K, Uemura T et al (2007) Self-organizing mechanism for development of space-filling neuronal dendrites. PLoS Comput Biol 3(11):e212

Tanaka G, Yamane T, Héroux JB et al (2019) Recent advances in physical reservoir computing: a review. Neural Netw 115:100–123

Tompris I, Chatzipaschalis IK, Chatzinikolaou TP, et al (2024) A reaction-diffusion cellular automata model for mycelium-based engineered living materials evolution. In: International conference on cellular automata for research and industry, Springer, pp 253–264

Tsakalos KA, Sirakoulis GC, Adamatzky A et al (2021) Protein structured reservoir computing for spike-based pattern recognition. IEEE Trans Parallel Distrib Syst 33(2):322–331

Tsompanas MA, Chatzinikolaou TP, Sirakoulis GC (2022) Cellular automata application on chemical computing logic circuits. In: Chopard B, Bandini S, Dennunzio A et al (eds) Cellular Automata. Springer, Cham, pp 3–14

Vaxevanellis E, Liolis O, Chatzinikolaou TP, et al (2024) Cellular automata-based system for traffic and parking management in urban centers. J Cell Automata 18

Vidal-Diez de Ulzurrun G, Baetens J, Van den Bulcke J et al (2017) Modelling three-dimensional fungal growth in response to environmental stimuli. J Theor Biol 414:35–49

Weimar JR (1997) Cellular automata for reaction-diffusion systems. Parallel Comput 23(11):1699–1715

Zhang Z (2018) Improved adam optimizer for deep neural networks. In: 2018 IEEE/ACM 26th international symposium on quality of service (IWQoS), IEEE, pp 1–2

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Authors and Affiliations

Ioannis Tompris[1] · Ioannis K. Chatzipaschalis[1,2] · Theodoros Panagiotis Chatzinikolaou[1] · Georgios Kleitsiotis[1] · Karolos-Alexandros Tsakalos[1] · Iosif-Angelos Fyrigos[1] · Michail-Antisthenis Tsompanas[3] · Andrew Adamatzky[3] · Phil Ayres[4] · Georgios Ch. Sirakoulis[1]

✉ Ioannis Tompris
itompris@ee.duth.gr

Ioannis K. Chatzipaschalis
ichatzip@ee.duth.gr

Theodoros Panagiotis Chatzinikolaou
tchatzin@ee.duth.gr

Georgios Kleitsiotis
gkleitsi@ee.duth.gr

Karolos-Alexandros Tsakalos
ktsakalo@ee.duth.gr

Iosif-Angelos Fyrigos
ifyrigos@ee.duth.gr

Michail-Antisthenis Tsompanas
antisthenis.tsompanas@uwe.ac.uk

Andrew Adamatzky
adamatzky@gmail.com

Phil Ayres
Phil.Ayres@kglakademi.dk

Georgios Ch. Sirakoulis
gsirak@ee.duth.gr

[1] Department of Electrical and Computer Engineering, Democritus University of Thrace, Xanthi, Greece

[2] Department of Electronic Engineering, Universitat Politécnica de Catalunya, Barcelona, Spain

[3] Department of Computer Science and Creative Technologies, University of the West of England, Bristol, UK

[4] Institute of Architecture and Technology, Royal Danish Academy, Copenhagen, Denmark